

Computational Economics

Lab: Introduction to Python

Development tools, scripts, notebooks, and reproducible results

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Run a Small Program and Check Its Result

Example: Convert a Temperature

You want to convert a temperature from Celsius to Fahrenheit.

- ▶ What information does the program need as input?
- ▶ Can you run the same calculation from a script and a notebook?
- ▶ If the input changes from 20 to 25 degrees Celsius, what should happen to the output?

Today's goal: run, explain, modify, and verify a small program in both file formats.

What Is an IDE?

- ▶ An **integrated development environment (IDE)** brings together tools for writing, running, and debugging programs.
- ▶ Typical features include a file explorer, code editor, completion suggestions, debugger, terminal, and version-control tools.
- ▶ We use **VS Code**: an extensible editor that provides an integrated Python workspace through its Python and Jupyter extensions.
- ▶ Other options include **PyCharm**, a Python-focused IDE, and **Cursor**, another code editor.

The editor helps you work with code; a separately selected Python interpreter executes it.

Find Your Way around VS Code

Part of the interface	What you use it for
Explorer	Open the project folder and find its files
Editor	Read and edit a script or notebook
Terminal	Run commands and inspect their output
Run and Debug	Execute code and inspect it step by step
Source Control	Review changes and create Git commits
Extensions	Add Python and Jupyter support

Open the project folder first. Then open a file and select its Python interpreter or notebook kernel.

Know Which Tool Does the Work

Tool or component	Role
Python	The language in which we express the calculation
Interpreter	The program that executes Python code
VS Code	An editor for files, projects, and notebooks
Package	Additional reusable code, such as NumPy or pandas
Environment	A project-specific Python setup and installed packages
Notebook kernel	The running process that executes notebook cells

Installing an editor does not install every interpreter or package. Check which Python your editor and notebook are using.

Two File Formats, One Calculation

Script: `.py`

Plain-text Python source

Run the file from the beginning

Useful for repeatable tasks

Check the printed result

Notebook: `.ipynb`

A document with code cells, text, and saved outputs

Run cells through a selected kernel

Useful for explanations and exploration

Restart the kernel and run all cells to verify

Use the supplied `temperature.py` and `temperature.ipynb`. Both should produce the same converted temperature.

Understand the Input, Calculation, and Output

Example: Celsius to Fahrenheit

$$F = \frac{9}{5}C + 32.$$

- ▶ **Input:** a temperature C in degrees Celsius.
- ▶ **Calculation:** multiply by $9/5$, then add 32.
- ▶ **Output:** the corresponding temperature F in degrees Fahrenheit.
- ▶ **Check:** $20 \times 9/5 + 32 = 68$. At 25 degrees Celsius, the result should be 77 degrees Fahrenheit.

The starter uses Python's standard library; no additional calculation package is required.

Set Up and Run the Script

1. Install a supported Python 3 release, VS Code, and Microsoft's Python and Jupyter extensions.
2. Open your `computational-economics` folder in VS Code. Copy the starter files into it.
3. Use **Python: Create Environment** to create a `.venv` environment, then select its interpreter.
4. Open `temperature.py` and choose **Run Python File in Terminal**.
5. Check the displayed interpreter path, Python version, and result: 20.0 C = 68.0 F.

Follow the linked setup guide for your operating system. The terminal displays commands and their output; the selected interpreter executes the script.

Run a Notebook from a Fresh Kernel

1. Open `temperature.ipynb` and select the Python environment you created.
2. Install notebook execution support, such as `ipykernel`, in that environment when prompted.
3. Read the explanation and run the cells in order.
4. Restart the kernel, run all cells, and check that the result is still 68 degrees Fahrenheit.

Example: An Old Output Can Be Misleading

A notebook can display a result saved during an earlier run. A fresh kernel clears stored variables; running all cells checks whether the saved code can recreate the result.

Read the Error and Check the Setup

What you observe	First check
Python is not found	Installation and the interpreter selected in VS Code
A package cannot be imported	Whether it is installed in that interpreter's environment
A notebook variable is undefined	Cell order; restart and run all cells
A file cannot be found	The project folder and the file path
The script runs but the answer is wrong	Inputs, calculation, units, and expected result

When asking for help, include the exact error, the relevant code, your operating system, and the selected interpreter. Check one possible cause at a time.

Homework: Run, Modify, and Explain

1. Install Python and set up your project environment.
2. Run the supplied `temperature.py`; record the Python version and baseline result.
3. Run `temperature.ipynb` from a fresh kernel and reproduce the baseline result.
4. Change the input from 20 to 25 degrees Celsius in both files. Predict and verify the new result: 77 degrees Fahrenheit.
5. Explain in your README what changed and why. Commit the code and notebook to the Git repository from the other lab, then push.

Completion evidence: both files run successfully, their results agree, the README describes how to run them, and the latest version is visible in your GitHub repository.

Record one check you performed to verify the result.

References and Further Reading

1. **Prof. Feng Zhigang (2026), AI for Economic Research: Dynamic Models, Language, and Agents. [Online course]**
 - ▶ Session 1.5: The Workbench: Python, VS Code, Git and GitHub
2. **Microsoft, Visual Studio Code Documentation. [Webpages]**
 - ▶ Getting Started with Python in VS Code
 - ▶ Jupyter Notebooks in VS Code
3. **Python Software Foundation, Python.org. [Webpage]**
 - ▶ Download Python